Target features and user journey map

What the Beta must let a keyman, patrolman and JE/SSE do on a foot patrol, step by step, and which of the 18 target features carries each step. Screens are rendered from the app's actual Compose code in CI. Hindi and English throughout; voice phrases shown in both.

3
user journeys: daily patrol, danger found, review and closure

18
target features, each traced to a Phase 1 requirement with an acceptance test

13
workflow stages from beat planning to audited history (L1–L5 deliverables)

67
defect classes offered in the Beta, bilingual, 14 of them protect-first

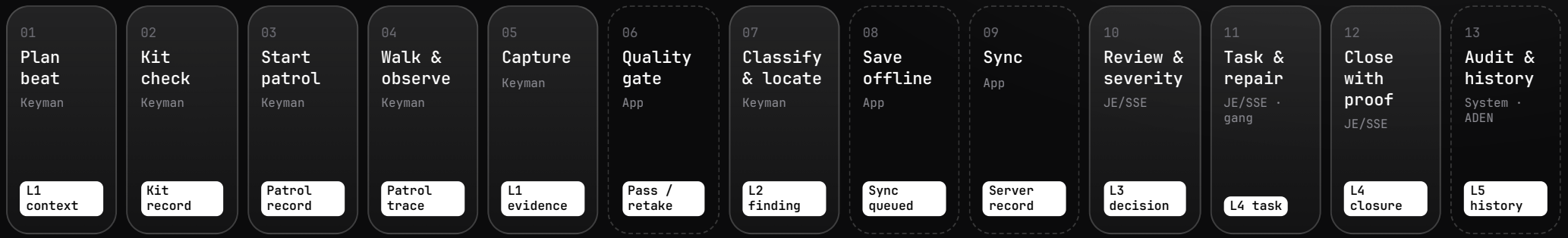
01 · Core workflow
Thirteen stages, who acts, what each stage hands on, and the protect-first branch.

02–04 · Journey maps
J1 daily patrol, J2 danger found, J3 review and closure: step, screen, interaction, deliverable, features.

05–06 · Target features
F01–F18 grouped by capability, with implementation, acceptance test and journey coverage.

One record, from the beat to an audited closure

Each stage hands a defined deliverable to the next. While walking, the keyman adds at most one tap or one spoken word per finding; classification happens only once stopped. Dashed stages are done by the app or server without user effort.



J1 Keyman daily patrol · stages 01–08

J2 Danger found · protect first, then 05–08

J3 JE/SSE review and closure · stages 09–13

PROTECT-FIRST BRANCH (BETWEEN 04 AND 05)

For the 14 protect-first classes (rail or weld fracture, washaway, obstruction and others), capture stays locked until the keyman confirms **line protected** and **Station Master informed**. Both are timestamped and stored with the finding. No voice command can report or skip it (rule R2).

RULES THE BACKEND ENFORCES

R1 Beta classes only

R2 Protect first

R3 AI = candidates only

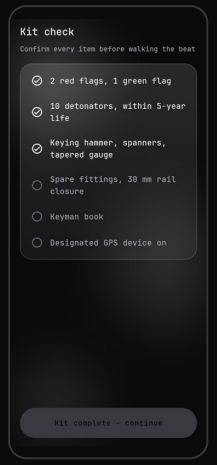
R4 Only JE/SSE/ADEN decide

R5 Closure needs proof

R6 Hash-chained audit

Keyman daily patrol (PT-01)

Walk the beat, record every defect with photo and location, hand a complete keyman book to the JE/SSE.

STEP	01 Open app	02 Kit check	03 Walk the beat	04 Spot missing ERC	05 Classify	06 Record action	07 Save	08 End patrol
SCREEN								
USER DOES	Sees today's beat, patrol type and sync state	Ticks six safety items	Watches track and trains; app stays silent	Captures without stopping	Confirms defect, km, line, rail once stopped	Notes what was done (e.g. ERC recouped)	Hears and feels the confirmation, walks on	Reviews auto-compiled book; submits
INTERACTION	Tap Start patrol · 'start patrol' / 'गश्त शुरू'	Continue unlocks only when all ticked	Live camera behind glass HUD; km in large figures	Tap Capture · 'capture' / 'फोटो लो'	Suggested chips · 'missing clip' / 'क्लिप गायब'	Optional dictated note · 'note …' / 'टिप्पणी …'	Haptic + spoken 'saved'; back to HUD	One Submit · 'end patrol' / 'गश्त समाप्त'
DELIVERABLE	–	Kit record	Patrol trace	L1 image + GNSS	L2 finding PW-FS-001	Keyman-book entry	Sync queued	Patrol report
FEATURES	F10 F16 Offline store and sync Language switch	F09 Kit check	F08 F17 Location Accessibility	F01 F02 F04 Evidence capture Image-quality gate Voice commands	F03 F04 F08 Bilingual defect taxonomy Voice commands Location	F05 Voice dictation of notes	F06 F10 Spoken and haptic confirmation Offline store and sync	F11 F10 Keyman-book summary Offline store and sync

Danger found (rail fracture, washaway)

Protect the line first, inform the Station Master, then record evidence with protection times.

STEP	01 Notice danger	02 Protect the line	03 Confirm protection	04 Capture evidence
SCREEN				
USER DOES	Stops; danger takes priority over recording	Follows on-screen protection steps	Ticks 'line protected' and 'SM informed'	Photographs the defect after protecting
INTERACTION	Tap Danger · 'danger' / 'खतरा' (opens guidance only)	Opaque sheet: flags, detonators 600 m and 1200 m (BG), inform SM	Deliberate taps only - no voice path	Capture unlocks after both ticks; class marked protect-first
DELIVERABLE	–	–	Escalation record + timestamps	L1/L2 with protection times
FEATURES	F04 F07 Voice commands · Protect-first guard	F07 Protect-first guard	F07 F14 Protect-first guard · Audit trail	F01 F02 F08 F10 Evidence capture · Image-quality gate · Location · Offline store and sync

JE/SSE review and closure

Decide severity from evidence without a site visit, get the repair done and close with proof.

STEP	01 Open inbox	02 Assess	03 Create task	04 Close
SCREEN				
USER DOES	Sees findings with km, defect, reporter, photo	Assigns RF-P1…P4; confirms or rejects	Raises work order for the gang	Closes only with closure photo
INTERACTION	Inbox sorted by protect-first, then age; duplicates grouped	Confirm enabled only after severity is set; AI candidates hidden (shadow)	Work-order reference; TMS export	Closure image must pass the quality gate (rule R5)
DELIVERABLE	–	L3 decision	L4 task	L4 closure + audit
FEATURES	F12 F18 Supervisor review · Duplicate suppression	F12 F15 Supervisor review · AI candidates (shadow)	F12 Supervisor review	F13 F02 F14 Closure with evidence · Image-quality gate · Audit trail

Eighteen features, each with a test that proves it

Filled dots show the journeys a feature serves. "Shadow" means the AI runs and stores candidates but users do not see them until field thresholds are met; "Server" runs in the backend only.

ID	FEATURE	IMPLEMENTATION (STANDARD LIBRARY FIRST)	ACCEPTANCE	TRACES TO	J1	J2	J3	STATUS
CAPTURE & EVIDENCE								
F01	**Evidence capture**	CameraX; JPEG to app storage; SHA-256 on upload	Capture ≤ 1 s from tap; works offline	REQ-03, REQ-14	●	●	○	Beta
F02	**Image-quality gate**	Device + server, same metrics; 'insufficient evidence' outcome	Thresholds in Ch. 13	REQ-06	●	●	●	Beta
F15	**AI candidates (shadow)**	ONNX detector behind quality gate; candidates stored, not shown until thresholds met	Feature flag	REQ-05	○	○	●	Shadow
CLASSIFY & LOCATE								
F03	**Bilingual defect taxonomy**	67 Beta classes, search in both scripts, protect-first and measurement tags	Every class has Hindi label (unit test)	D02	●	○	○	Beta
F08	**Location**	Fused Location + km/mast entry + OCR assist (ML Kit Devanagari)	Inspector confirms	REQ-03	●	●	○	Beta
F18	**Duplicate suppression**	GNSS ≤ 25 m + embedding similarity (server)	Benchmark Ch. 11	—	○	○	●	Server
VOICE & FEEDBACK								
F04	**Voice commands**	On-device SpeechRecognizer hi-IN / en-IN + closed vocabulary matcher	Intent accuracy thresholds (Ch. 13)	REQ-10	●	●	○	Beta
F05	**Voice dictation of notes**	Same recogniser; note language stored	Note saved with language tag	PR-07	●	○	○	Beta
F06	**Spoken and haptic confirmation**	TextToSpeech + HapticFeedback CONFIRM	Toggle in Settings	REQ-10	●	○	○	Beta

Record, review and access

Thresholds referenced as Ch. 13 and benchmark results as Ch. 11 are in the Solution Design Report.

ID	FEATURE	IMPLEMENTATION (STANDARD LIBRARY FIRST)	ACCEPTANCE	TRACES TO	J1	J2	J3	STATUS
SAFETY								
F07	**Protect-first guard**	Sheet + two confirmations; API rule R2; no voice path	Unit + API tests	REQ-01	○	●	○	Beta
F09	**Kit check**	Checklist before patrol	All items required	REQ-11	●	○	○	Beta
RECORD & SYNC								
F10	**Offline store and sync**	Room + WorkManager (network constraint, exponential backoff)	No data loss on airplane-mode test	REQ-02	●	●	○	Beta
F11	**Keyman-book summary**	Auto-compiled from findings (CAMTECH 2021 B17 fields)	Matches saved findings	PR-07	●	○	○	Beta
F14	**Audit trail**	Append-only SHA-256 hash chain; /audit/verify	Chain verified in tests	REQ-14	○	●	●	Beta
SUPERVISOR								
F12	**Supervisor review**	Inbox; severity; confirm/reject; API R4	Keyman cannot review (API test)	REQ-05	○	○	●	Beta
F13	**Closure with evidence**	Closure evidence must pass quality gate; API R5	API test	REQ-09	○	○	●	Beta
LANGUAGE & ACCESS								
F16	**Language switch**	Per-app language (Android 13 system setting + in-app)	Hindi screens rendered in CI	—	●	○	○	Beta
F17	**Accessibility**	Large-text mode, ≥ 48 dp targets (HUD 60–88 dp), TalkBack labels, greyscale contrast	Stage V6	—	●	○	○	Beta